Chapter # 1. 1st Year
Basic Concepts of Information Technology

Q. No. 1. What is computer Science and Information Technology?

Computer Science

Computer Science is the Study of computers, their design (*see* computer architecture), and their uses for computation, data processing, and systems control, including design and development of computer hardware and software and programming. The field encompasses theory, mathematical activities such as design and analysis of algorithms, performance studies of systems and their components, and estimation of reliability and availability of systems by probabilistic techniques.

Information Technology

Stands for "Information Technology," and is pronounced "I.T." It refers to anything related to “IFORMATION” using computing technology and other technology such as networking, hardware, software, the Internet, or the people that work with these technologies. Many companies now have IT departments for managing the computers, networks, and other technical areas of their businesses. IT jobs include computer programming, network administration, computer engineering, Web development, technical support, and many other related occupations. Since we live in the "information age," information technology has become a part of our everyday lives. That means the term "IT," already highly overused, is here to stay.

Q. No. 2. What are the characteristics of a Computer?

1) Speed:

A computer is a very fast device. It can perform in few seconds, the amount of work that a human being can do in an entire year. The speed of a computer measure in microseconds(10-6), nanoseconds(10-9) and picoseconds(10-12). A powerful computer is capable of performing several billion arithmetic operation per second.

2) Accuracy:

Computer is very accurate device. The accuracy of a computer is principle high. The degree of accuracy of a particular computer depends upon it’s design. Computer done very calculation with same accuracy. Errors can occur in a computer, However, these problem are mainly due to human rather than technological weaknesses.

3) Automatic(Automation):

A machine is no sooner than done to be automatic if it, works by itself without human intervention. Computer are automatic machines, because one started on a job, they carry on, until the job is finished, without any human assistance. However, computer

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being machines can not start them selves. They cannot go out find their own problems and solutions. They have to be instructed.

4) Diligence:

Computer has unlike human beings. A computer is free from dullness and lack of concentration. It can continuously work for hours without creating any error human beings in doing regular types of jobs, which requires great accuracy. Computer can perform the number of calculation with exactly the same accuracy and speed as the first one.

5) Versatility:

Versatility is a most important characteristic of computer. It means the capacity to perform completely different type of work. You may use your computer to prepare payroll slips. Next moment you may use it for inventory management or to prepare electric bills. A computer is capable of performing almost any task, if the task can be reduced to series of logical steps.

6) Memory(Power of Remembering):

Computer has a brain, but unlike human (Man) beings. A computer can store data and information. And recalled as long as you require it, for any numbers of years. Because of it's secondary storage capability. A data and information can be retrieved as long as desired by the user and can be recalled, as when required. The information recalled would be as accurate as on the day when it was fed to the computer.

7) No. I.Q.(Intelligence Quotient):

A computer is not a magical device. It is no intelligence of it's own. User can determine what tasks the computer will perform. So a computer cannot take its own decision as you can. (A computer can not take it own decision in this regard.).

8) No Feelings:

Computer has no feelings. They have no emotions, because they are machine. Computer is not like human heart and soul. Computer cannot make such judgments on their own. Their judgment is based on the instructions given to them in the from of programs, that are written by us.

Q. No. 3. What meant by computer system?

Computer System

A system is a group of integrated parts that have the common purpose of achieving the particular tasks.

Q. No. 4. What are parts of the computer system?

Parts of Computer System

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Hardware

The physical or tangible components that make up the computer are called hardware. Hardware means any part of the computer that you can touch or feel.

Software

Instructions that tell a computer what to do. Software is the entire set of programs, procedures, and routines associated with the operation of a computer system.

Data

Data is a collection of facts, such as values or measurements.
It can be numbers, words, measurements, observations or even just descriptions of things
Data (plural of "**datum**") are typically the results of measurements and can be the basis of graphs, images, or observations of a set of variables. Data are often viewed as the lowest level of abstraction from which information and then knowledge are derived. *Raw data*, i.e. Unprocessed data, refers to a collection of numbers, characters, images or other outputs from devices that collect information to convert physical quantities into symbols.

Procedures

Procedures ate the steps that you must follow to accomplish a specific computer related task. Manuals containing rules and guidelines to follow when using software, hardware and data are available.

People

People are the computer operator also known as users or some time end user. People operate the hardware create the software, put the data into the system, and use the information that comes out of it.

Q. No. 5. Define Input device?

Input Device

An **input device** is any peripheral (piece of computer hardware equipment) used to provide data and control signals to an information processing system (such as a computer). Input and output devices make up the hardware interface

Q. No. 6. Discuss the Input devices with their functions?

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Keyboard Input Devices

In computers, the keyboard is the most basic input device of the modern computer, essential for user interfacing with the machine.

Consists of a number of "keys" (very similar to buttons in practice) typically with glyph representation for each character of the native alphabet, numbers, basic punctuation, mathematical symbols, brackets and several modifier keys to expand functionality.

Mouse

A device that controls the movement of the cursor or pointer on a display screen. A mouse is a small object you can roll along a hard, flat surface. Its name is derived from its shape, which looks a bit like a mouse, its connecting wire that one can imagine to be the mouse's tail, and the fact that one must make it scurry along a surface. As you move the mouse, the pointer on the display screen moves in the same direction. Mice contain at least one button and sometimes as many as three, which have different functions depending on what program is running. Some newer mice also include a *scroll wheel* for scrolling through long documents.

Trackball

Essentially, a trackball is a mouse lying on its back. To move the pointer, you rotate the ball with your thumb, your fingers, or the palm of your hand. There are usually one to three buttons next to the ball, which you use just like mouse buttons.

Scanner

In computing, an image scanner—often abbreviated to just scanner— is a device that optically scans images, printed text, handwriting, or an object, and converts it to a digital image. Common examples found in offices are variations of the *desktop scanner*

Hand-Held Scanner

A scanner that is moved by hand over the material being captured. Handheld scanners are small and less expensive than their desktop counterparts but partially rely on the user's to move the unit across the paper. Trays are available that keep the scanner moving in a straight line

Flatbed Scanner

A scanner that provides a flat, glass surface to hold pages of paper, books and other objects for scanning. The scan head is moved under the glass across the page. Sheet feeders are usually optionally available that allow multiple sheets to be fed automatically.

Sheet-Fed Scanner

A scanner that allows only paper to be scanned rather than books or other thick objects. It moves the paper across a stationary scan head.

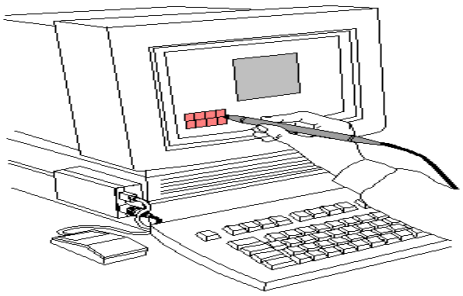
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Joystick

Computer input device, especially helpful when playing computer games. It consists of a handle that can be pointed in different directions. Because the computer can sense which direction the joystick is pointed, it can be used to control the movements of objects displayed on the computer screen.

Light Pen

A **light pen** is a computer input device in the form of a light-sensitive wand used in conjunction with a computer's CRT TV set or monitor. It allows the user to point to displayed objects, or draw on the screen, in a similar way to a touch screen but with greater positional accuracy. It was long thought that a light pen can work with any CRT-based display, but not with LCD screens



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Microphone

A microphone is a device that converts mechanical energy waves or sound into electrical energy waves. Speaking into a microphone excites (moves) a diaphragm that is coupled to a device that creates an electrical current proportional to the sound waves produced.

Digital Camera

A camera that stores images digitally rather than recording them on film. Once a picture has been taken, it can be downloaded to a computer system, and then manipulated with a graphics program and printed

Q. No. 7. Define Output Device?

Output Devices

An **output device** is any piece of computer hardware equipment used to communicate the results of data processing carried out by an information processing system (such as a computer) to the outside world.

Q. No. 8. Define Monitor? Also define its types?

Monitors

A **monitor** or **display** (sometimes called a **visual display unit**) is an electronic visual display for computers. The monitor comprises the display device,

There are basic two types of Monitors.

CRT (Cathode Ray Tube)

CRT monitors, which is similar to the television screen and also works in the same way. It uses a large vacuum tube called Cathode Ray Tube(CRT).

Flat – Panel

A flat panel monitor can refer to either a computer or television monitor that does not use cathode ray tube (CRT) technology, but commonly LCD or plasma technology. This allows the monitor to have a thin profile, which is how the flat panel monitor gets its name. Because of its light weight, small footprint, clarity and digital technology

Liquid crystal Display or LCD, which is usually used with portable computers but now has become very popular with the desktop computers too.

Q. No. 8. What are the main factors to determine the monitor's qualities?

Factors to determine a monitor's qualities.

1) Resolution

The resolution of a monitor indicates how density packed the pixels are. In general resolution provides a shaper image.

2) Bandwidth

The range of signal frequencies the monitor can handle this determine how much data it can process and therefore how fast it can refresh at higher resolutions.

3) Refresh Rate

How many times per second the screen is refreshed (redrawn). To avoid flickering the refresh should be at least 72 Hz.

4) Dot Pitch

The amount of space between each pixel. The smaller the dot pitch, the sharper the image.

5) Convergence

The clarity and sharpness of each pixel.

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Q. No. 9. What is printer? Also define its types?

Printers

Printer is a device that accepts text and graphic output from a computer and transfers the information to paper, usually to standard size sheets of paper. Printers are sometimes sold with computers, but more frequently are purchased separately. Printers vary in size, speed, sophistication, and cost. In general, more expensive printers are used for higher-resolution color printing.

Printer fall into basic two categories.

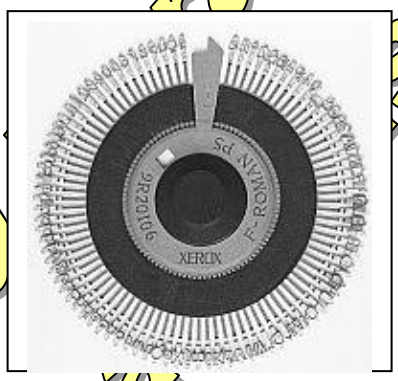
a) Impact Printer

Impact Printers include all printers that works by striking an Ink ribbon. Daisy wheel, dot matrix and line printers are impact printer. Impact printer are much noisier than other printer.

Daisy – Wheel

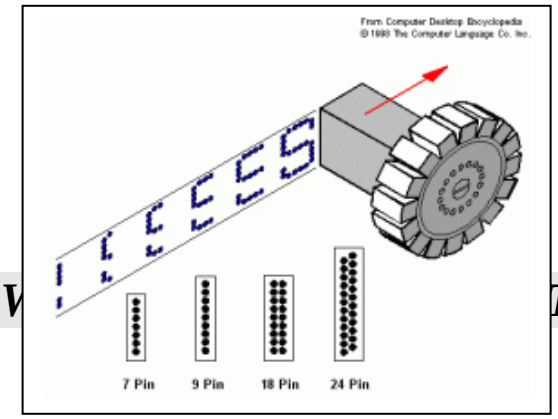
The print head of a daisy wheel printer is a circular wheel, about 3 inches in diameter with arms or spokes as show in the image bellow. The character shapes are embossed at the outer end of the arms. The shape of printer wheel resembles the petals of a daisy flower and hence, it is named daisy wheel printer.

To print a character, the daisy wheel rotates so that he required character is positioned just in front of the printer ribbon.



Dot – Matrix

A **dot matrix printer** or **impact matrix printer** is a type of computer printer with a print head that runs back and forth, or in an up and down motion, on the page and prints by impact, striking an ink-soaked cloth ribbon against the paper, much like the print mechanism on a typewriter. However, unlike a typewriter or daisy wheel printer, letters are drawn out of a dot matrix, and thus, varied fonts and arbitrary graphics can be produced. Because the printing involves mechanical pressure, these printers can create carbon copies and carbonless copies.



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Line Printer

A line printer contains a chain of character or pins that print an entire line at a one time. it produce low quality print.

b) Non-impact Printer

Non-impact printers include laser printer, ink jet printer etc.

Ink – jet

An **inkjet printer** is a type of computer printer that creates a digital image by propelling droplets of ink onto paper. Inkjet printers are the most commonly used type of printer and range from small inexpensive consumer models to very large professional machines

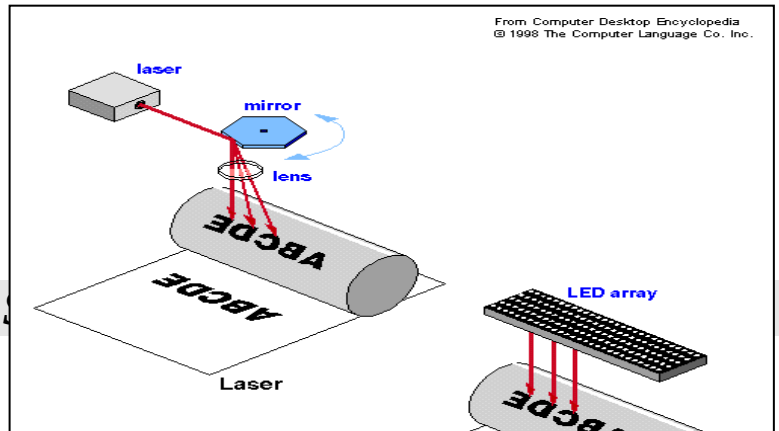
Laser Printer

A type of printer that utilizes a laser beam to produce an image on a drum. The light of the laser alters the electrical charge on the drum wherever it hits. The drum is then rolled through a reservoir of toner, which is picked up by the charged portions of the drum. Finally, the toner is transferred to the paper through a combination of heat and pressure. This is also the way copy machines work.

LCD & LED

An electro photographic printer that uses a matrix of LEDs as its light source. The LED mechanism is much simpler than its laser printer counterpart. A stationary array of LEDs is used instead of numerous moving parts, and the LEDs are selectively beamed onto the drum. LED printers are available from very low-end 4 ppm personal printers to huge digital printing presses that can print more than 700 ppm

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Thermal Printer

An inexpensive printer that works by pushing heated pins against heat sensitive paper. Thermal printers are widely used in calculator and fax machines.

Q. No. 10. What is Plotter?

Plotters

Plotter is a special kind of out put device that print graphic on papers, which includes drawing, graphic, making maps, plotting engineering drawings and machine components.

Q. No. 11. Define the types of Plotters?

Types of plotter

Drum Plotter

A type of pen plotter that wraps the paper around a drum with a pin feed attachment. The drum turns to produce one direction of the plot, and the pens move to provide the other. The plotter was the first output device to print graphics and large engineering drawings

Flatbed Plotter

A graphics plotter that contains a flat surface that the paper is placed on. The size of this surface (bed) determines the maximum size of the drawing.

Electrostatic Plotter

A plotter that uses an electrostatic method of printing. Liquid toner models use a positively charged toner that is attracted to paper which is negatively charged by passing by a line of electrodes (tiny wires or nibs). Models print in black and white or color, and some handle paper up to six feet wide. Newer electrostatic

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plotters are really large-format laser printers and focus light onto a charged drum using lasers or LEDs.

Q. No. 12. Define the tern Softcopy and Hardcopy?

Concept of Softcopy and Hardcopy

Softcopy

A **soft copy** is the unprinted digital document file. This term is often contrasted with **hard copy**. It can usually be viewed through an appropriate editing program, such as word processing programs, database programs, or presentation software, depending on the file type.

It can be transported from one computer to another through file transfer/downloading mechanisms such as ftp or http, as an email attachment, or through USB drives and other disk drives. Keeping a digital copy of a document can allow easy editing of it later on.

Hardcopy

A printout of data stored in a computer. It is considered *hard* because it exists physically on paper, whereas a *soft* copy exists only electronically.

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Q. No. 13. What are the components of System Unit?

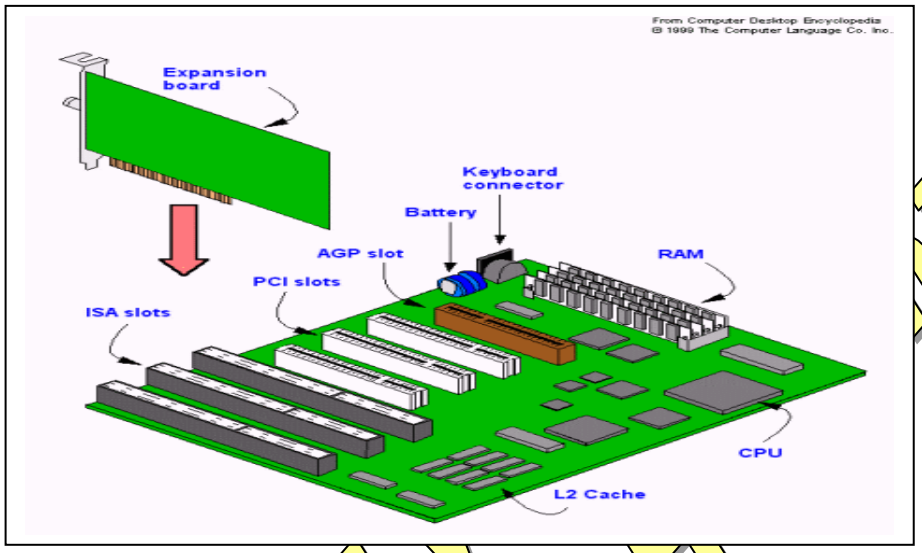
These are the components which formed a system unit.

1. The power supply
2. The motherboard
3. Microprocessor (including register, bus, clock and coprocessor)
4. Main memory of random access memory (RAM)
5. Read only memory (RAM)
6. Other forms of Memory (cache, VRAM, flash)
7. Expansion cards and slots
8. Ports

Q. No. 14. Define the motherboard?

A motherboard is the central printed circuit board (PCB) in many modern computers and holds many of the crucial components of the system, providing connectors for other peripherals. The motherboard is sometimes alternatively known as the main board, system board

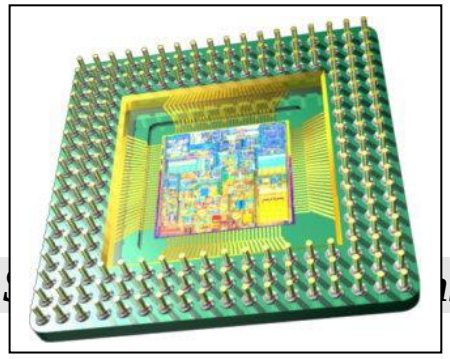
In a modern desktop computer, the motherboard contains the CPU and memory sockets as well as the chipset, which houses the control circuits for all the peripheral devices. It also has a PCI-Express slot for a high-end display adapter and PCI slots for additional peripherals. Laptop motherboards have no expansion slots for more peripherals



Q. No. 15. Define the Microprocessor with their parts and functions?

A microprocessor, sometimes called a *logic chip*, is a computer processor on a microchip.

The microprocessor contains all, or most of, the central processing unit (CPU) functions and is the "engine" that goes into motion when turn computer on. A microprocessor is designed to perform arithmetic and logic operations that make use of small number-holding areas called *registers*. Typical microprocessor operations include adding, subtracting, comparing two numbers, and fetching numbers from one area to another. These operations are the result of a set of instructions that are part of the microprocessor design.



Parts of Microprocessor
Control Unit (CU)
Arithmetic Logic Unit (ALU)

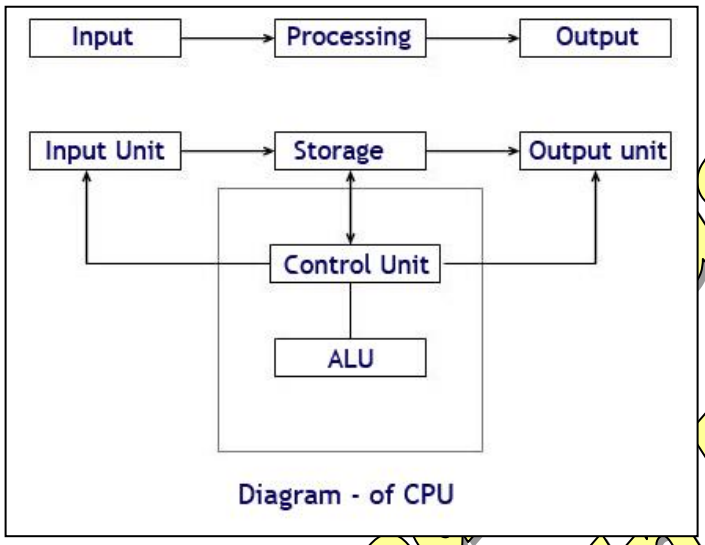


Diagram - of CPU

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Control Unit (CU)

The control unit is the circuitry that controls the flow of data through the processor, and coordinates the activities of the other units within it. In a way, it is the "brain within the brain", as it controls what happens inside the processor, which in turn controls the rest of the PC.

Arithmetic Logic Unit (ALU)

(**A**rithmetic **L**ogic **U**nit) The high-speed CPU circuit that does calculating and comparing. Numbers are transferred from memory into the ALU for calculation, and the results are sent back into memory. Alphanumeric data are sent from memory into the ALU for comparing. The results are tested by GOTOs

OR

An arithmetic-logic unit (ALU) is the part of a computer processor (CPU) that carries out arithmetic and logic operations on the operands in computer instruction words. In some processors, the ALU is divided into two units, an arithmetic unit (AU) and a logic unit (LU). Some processors contain more than one AU - for example, one for *fixed-point* operations and another for *floating-point* operations. (In personal computers floating point operations are sometimes done by a floating point unit on a separate chip called a numeric coprocessor.)

Features of microprocessor

1. Register and word size

2. Buses and bus width
3. Clock and clock speed
4. Coprocessor

Q. No. 16. Define the Register and word size?

A register is one of a small set of data holding places that are part of a computer processor . A register may hold a computer instruction , a storage address, or any kind of data (such as a bit sequence or individual characters). Some instructions specify registers as part of the instruction. For example, an instruction may specify that the contents of two defined registers be added together and then placed in a specified register. A register must be large enough to hold an instruction - for example, in a 32-bit instruction computer, a register must be 32 bits in length. In some computer designs, there are smaller registers - for example, *half-registers* - for shorter instructions. Depending on the processor design and language rules, registers may be numbered or have arbitrary names.

Q. No. 17. What is Bus and Bus width?

Buses

A set of parallel conductors, which allow devices attached to it to communicate with the CPU.

When referring to a computer, the **bus** also known as the **address bus, data bus,** or **local bus** is a data connection between two or more devices connected to the computer. For example, a bus enables a computer processor to communicate with the memory or a video card to communicate with the memory.

A bus is capable of being parallel or a serial bus and today all computers utilize two bus types, an internal or local bus and an external bus. An internal bus enables a communication between internal components such as a computer video card and memory and an external bus is capable of communicating with external components such as a USB or SCSI device.

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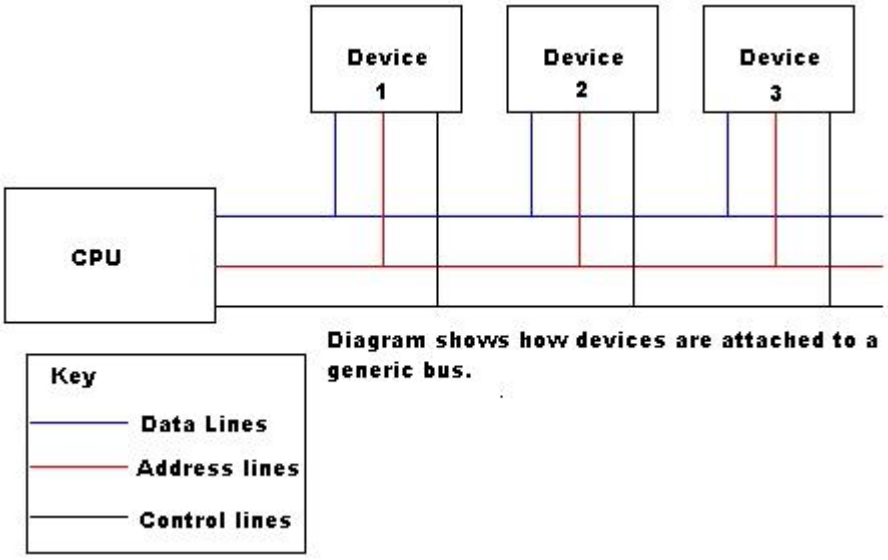


Diagram shows how devices are attached to a generic bus.

Q. No. 18. Discuss the types of BUS?

The bus consists of three main parts:

Address Bus

The address bus is used by the processor in a computer to locate a piece of data from the RAM (Random Access Memory) that it needs to access. No real data is carried via the address bus, as this is the job of the

Data Bus.

The pieces of data that the address bus locates are called "addresses" and the address bus transfers "data addresses" to and from the control unit.

Control Bus

A **control bus** is (part of) a computer bus, used by CPUs for communicating with other devices within the computer. While the address bus carries the information on which device the CPU is communicating with and the data bus carries the actual data being processed, the control bus carries commands from the CPU and returns status signals from the devices

Q. No. 19. What is Clock also define clock speed?

The clock control how fast all the operation within a CPU takes place.
clock speed refers to the number of pulses per second generated by an oscillator that sets the tempo for the processor. Clock speed is usually measured in MHz (megahertz, or millions of pulses per second) or GHz (gigahertz, or billions of pulses per second).

Q. No. 20 What is the function of Coprocessor?

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A **coprocessor** is a computer processor used to supplement the functions of the primary processor (the CPU).
A coprocessor may not be a general-purpose processor in its own right. Some coprocessors cannot fetch instructions from memory, execute program flow control instructions, do input/output operations, manage memory, and so on. These processors require the host main processor to fetch the coprocessor instructions and handle all other operations aside from the coprocessor functions. In some architecture the coprocessor is a more general-purpose computer, but carries out only a limited range of functions under the close control of a supervisory processor.

Q. No. 21. Define the Main Memory or Random Access Memory?

Random-access memory (**RAM**) is a form of computer data storage. Today, it takes the form of integrated circuits that allow stored data to be accessed.
RAM holds instructions and data. When the computer is switched off, the contents of the RAM are lost. It's just like a calculator- switch that off and you lose the contents.

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There are two types of RAM

1- **SRAM (Static RAM)**
2- **DRAM (Dynamic RAM)**

There are two different types of RAM: DRAM (Dynamic Random Access Memory) and SRAM (Static Random Access Memory). The two types differ in the technology they use to hold data, with DRAM being the more common type. In terms of speed, SRAM is faster. DRAM needs to be refreshed thousands of times per second while SRAM does not need to be refreshed, which is what makes it faster than DRAM. DRAM supports access times of about 60 nanoseconds, SRAM can give access times as low as 10 nanoseconds. Despite SRAM being faster, it's not as commonly used as DRAM because it's so much more expensive. Both types of RAM are *volatile*, meaning that they lose their contents when the power is turned off.

Q. No. 22. Define the ROM (Read only Memory)

ROM (Read only Memory) is "built-in" computer memory containing data that normally can only be read, not written to. ROM contains the programming that allows your computer to be "booted up" or regenerated each time you turn it on. Unlike a computer's random access memory (RAM), the data in ROM is not lost when the computer power is turned off. The ROM is sustained by a small long-life battery in your computer.
If you ever do the hardware setup procedure with your computer, you effectively will be writing to ROM. It has special programs called Firmware.

Q. No. 23. What are the types of ROM?
There are several types of ROM

PROM (Programmable Read-Only Memory)

*Programmable **r**ead-**o**nly **m**emory,* a memory chip on which data can be written only once. Once a program has been written onto a PROM, it remains there forever. Unlike RAM, PROMs retain their contents when the computer is turned off.

EPROM (Erasable Programmable Read-Only Memory)

EPROM (erasable programmable read-only memory) is programmable read-only memory (programmable ROM) that can be erased and re-used. Erasure is caused by shining an intense ultraviolet light through a window that is designed into the memory chip. (Although ordinary room lighting does not contain enough ultraviolet light to cause erasure, bright sunlight can cause erasure. For this reason, the window is usually covered with a label when not installed in the computer.)

EEPROM (Electrically Erasable Programmable Read-Only Memory)

*Electrically **e**rasable **p**rogrammable **r**ead-**o**nly **m**emory*. EEPROM is a special type of PROM that can be erased by exposing it to an electrical charge. Like other types of PROM, EEPROM retains its contents even when the power is turned off. Also like other types of ROM, EEPROM is not as fast as RAM.

Q. No. 24. What is the purpose of Expansion Cards and Slot?

The **expansion card** (also *expansion board*, *adapter card* or *accessory card*) in computing is a printed circuit board that can be inserted into an **expansion slot** of a computer motherboard or backplane to add functionality to a computer system via the expansion bus.

Expansion slots are specific slots on a PC motherboard that facilitate the placement of expansion cards.

Q. No. 25. What is Port?

Ports

On computer and telecommunication devices, a *port* (noun) is generally a specific place for being physically connected to some other device, usually with a socket and plug of some kind. Typically, a personal computer is provided with one or more serial ports and usually one parallel port.

Q. No. 26. Define the types of Ports?

Serial Ports?

A socket on a computer used to connect a modem, data acquisition terminal or other device via a serial interface (one data bit following the other). Serial ports provide very slow speeds and have been superseded by USB and other faster interfaces for peripheral connections to desktop computers. Although still widely used in data acquisition, the serial port is no longer found on new computers. Earlier PCs used the port for the mouse.

A serial port also called a male connector has 9 or 35 pins.

Parallel ports

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A socket on a computer for transmitting data in parallel, which means more then one bit transfer at a time. There are may be 8,16, or 32 channels, each channel carries one bit of information, so eight channels would be used to transmit 8 bits at a time.
It is also known as a **printer port**

USB ports

A USB port is a standard cable connection interface on personal computers and consumer electronics. USB ports allow stand-alone electronic devices to be connected via cables to a computer
U*niversal* ***S****erial* ***B****us,* an external bus standard that supports data transfer rates of 12 Mbps. A single USB port can be used to connect up to 127 peripheral devices, such as mice, modems, and keyboards. USB also supports *Plug-and-Play* installation and *hot plugging*.

Q. No. 27. Define Storage Devices?
In a computer, storage is the place where data is held in an electromagnetic or optical form for access by a computer processor.
Usually the term memory identifies primary data storage that comes in the form of chips, and the word storage is used for memory that exists on types of diskettes, secondary memory.

There are two types of Storage Devices
1- **Primary / Main Memory / Internal Memory**
2- **Secondary / Auxiliary Memory**

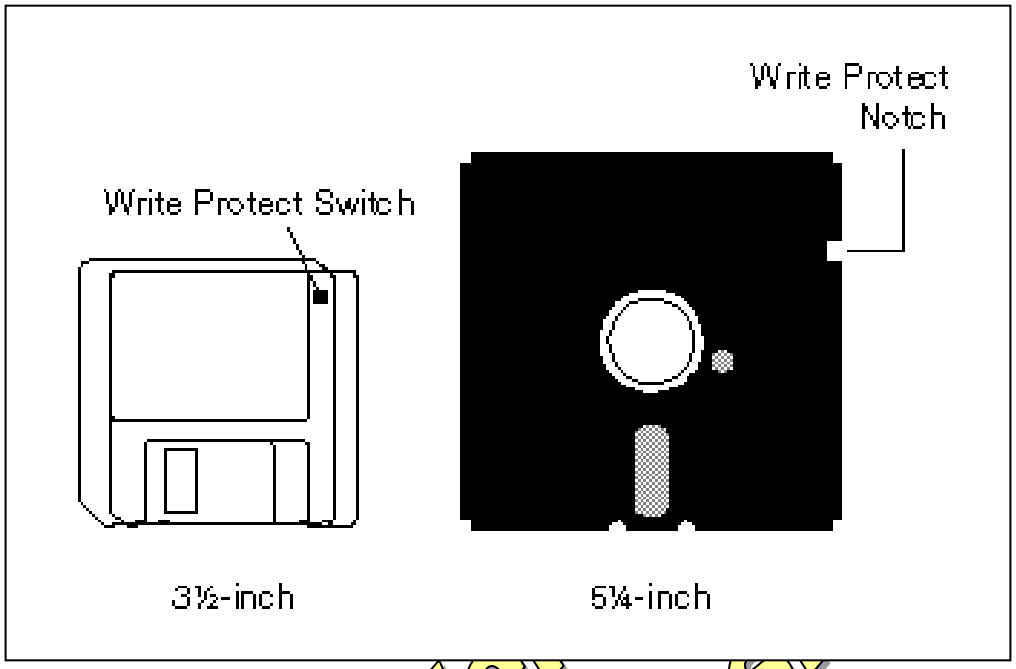
Q. No. 28. Define Primary / Main Memory of computer system?

Primary storage (or **main memory** or **internal memory**), often referred to simply as **memory**, is the only one directly accessible to the CPU. The CPU continuously reads instructions stored there and executes them as required. Any data actively operated on is also stored there in uniform manner.
Primary storage is much faster to access than secondary storage because of the proximity of the storage to the processor or because of the nature of the storage devices.

Q. No. 29. Define Floppy Diskette?

A soft magnetic disk. It is called *floppy* because it flops if you wave it (at least, the 5??-inch variety does). Unlike most hard disks, floppy disks (often called *floppies* or *diskettes*) are portable, because you can remove them from a disk drive. Disk drives for floppy disks are called *floppy drives.* Floppy disks are slower to access than hard disks and have less storage capacity, but they are much less expensive. And most importantly, they are portable.

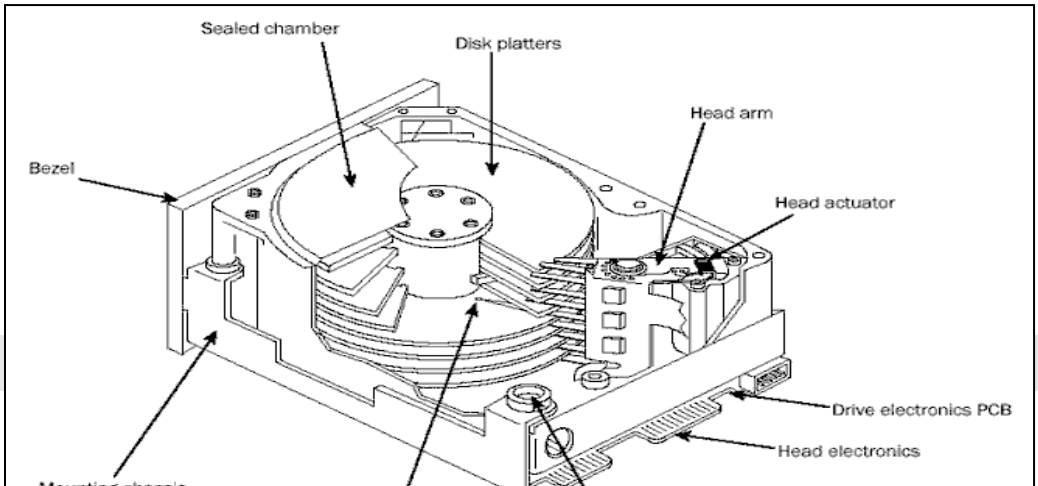
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Q. No. 30. Define Hard Disk?

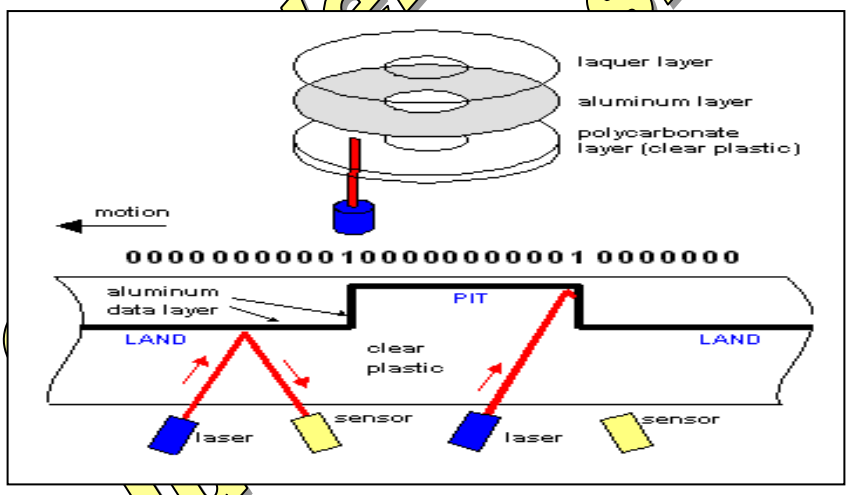
Hard disk, Magnetic storage medium for a microcomputer. Hard disks are flat, circular plates made of aluminum or glass and coated with a magnetic material. Hard disks for personal computers can store up to several gigabytes (billions of bytes) of information. Data are stored on their surfaces in concentric tracks. A small electromagnet, called a magnetic head, writes a binary digit (1 or 0) by magnetizing tiny spots on the spinning disk in different directions and reads digits by detecting the magnetization direction of the spots. A computer's hard drive is a device consisting of several hard disks, read/write heads,

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Q. No. 31. Define CD-ROM?

CD-ROM Stands for "Compact Disc Read-Only Memory." A CD-ROM is a CD that can be read by a computer with an optical drive. The "ROM" part of the term means the data on the disc is "read-only," or cannot altered or erased. Because of this feature and their large capacity, CD-ROMs are a great media format for retail software. The first CD-ROMs could hold about 600 MB of data, but now they can hold up to 700 MB. CD-ROMs share the same technology as audio CDs, but they are formatted differently, allowing them to store many types of data.



Q. No. 32. Define the Magnetic Tape?

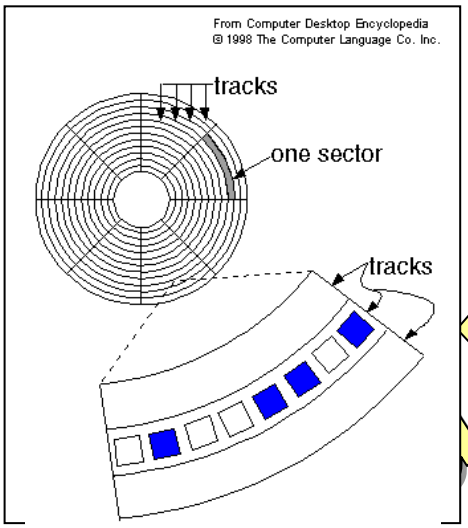
A sequential storage medium used for data collection, backup and archiving. Like videotape, computer tape is made of flexible plastic with one side coated with a ferromagnetic material. Tapes were originally open reels, but were superseded by cartridges and cassettes of many sizes and shapes.

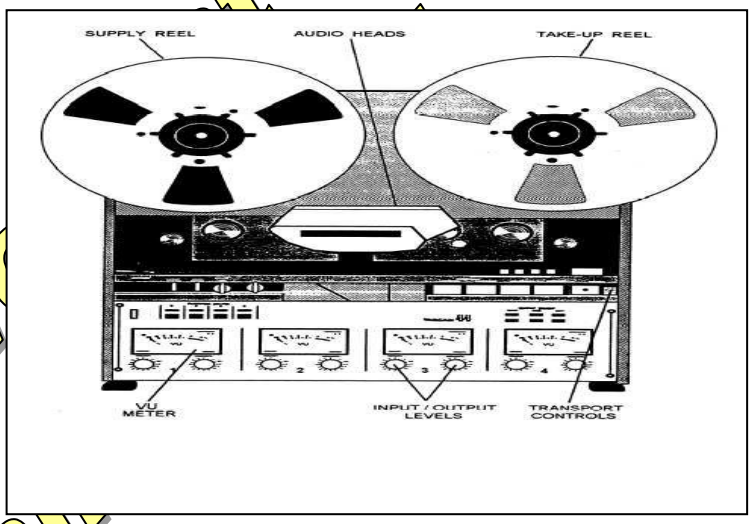
The major drawback of tape is its sequential format. Locating a specific record requires reading every record in front of it or searching for markers that identify predefined partitions

Tracks run parallel to the edge of the tape (linear recording) or diagonally (helical scan). A linear variation is serpentine recording, in which the tracks "snake" back and forth from the end of the tape to the beginning.

1. **Interblock gap**
2. **Interrecord gap**

Alternatively referred to as an **Interblock gap** and abbreviated as **IBG** or **IRG**, an **Interrecord gap** is a break between data records on hard drive and magnetic tape that serves as a safety margin for data overwrites. Interrecord gaps are also used as a marker for the end of data.





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Q. No. 33. What is Software?

Computer software is a collection of computer programs and related data that provide the instructions for telling a computer what to do and how to do it. In other words, software is a conceptual entity which is a set of computer programs, procedures, and associated documentation concerned with the operation of a data processing system. We can also say software refers to one or more computer programs and data held in the storage of the computer for some purposes. In other words software is a set of **programs,**

Q. No. 34. Define the types of Software?

There are two basic types of Software.

System Software

Application Software

System Software

System software is a program that manages and supports the computer resources and operations of a computer system while it executes various tasks such as processing data and information, controlling hardware components, and allowing users to use application

software. That is, systems software functions as a *bridge* between computer system hardware and the application software. System software is made up of many control programs, including the operating system, communications software and database manager

Application Software

Application software consists of Programs that direct computers to perform specific information processing activities for end users. These programs are called *application* packages because they direct the processing required for a particular use

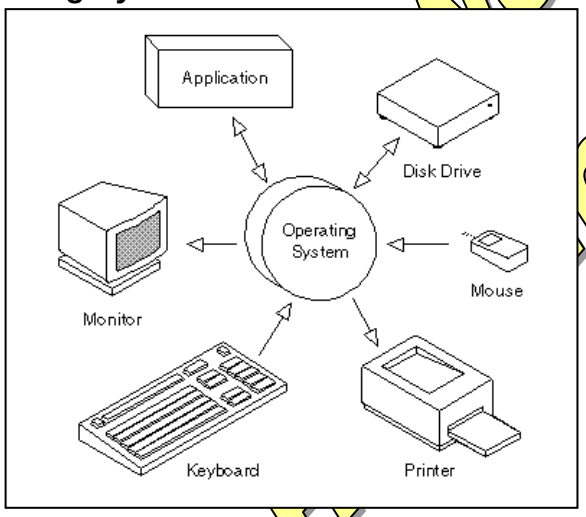
Q. No. 35. What is an Operating system?

Operating Systems

An operating system is a collection of integrated computer programs that provide recurring services to other programs or to the user of a computer. These services consist of disk and file management, memory management, and device management. In other words, it manages CPU operations, input/output activities, storage resources, diverse support services, and controls various devices.

Q. No. 36. Define Operating System Function?

Operating System Functions



An operating system executes many functions to operate computer system efficiently. Among them, four essential functions are the followings.

- **Resource Management:** An operating system manages a collection of computer hardware resources by using a variety of programs. It manages computer system resources, including its CPU, primary memory, *virtual memory*, secondary storage devices, input/output peripherals, and other devices.
- **Task Management:** The function of the operating system that controls the running of many tasks. It manages one program or many programs within a computer system simultaneously. That is, this function of operating system manages the completion of users' tasks. A task management program in an operating system provides each task and interrupts the CPU operations to manage tasks efficiently. Task management may involve a *multitasking* capability.
- **File management:** This is a function that manages data files. An operating system contains file management programs that provide the ability to create, delete, enter, change, ask, and access of files of data. They also produce reports on a file.

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- **User Interface:** It is a function of an operating system that allows users to interact with a computer. A user interface program may include a combination of menus, screen design, keyboard commands. A well-designed user interface is essential for an operating system to be popular. Because of the function, users can load programs, access files, and accomplish other tasks.

Q. No. 37. Define Operating System Environment?

- **Single-user, single task** - As the name implies, this operating system is designed to manage the computer so that one user can effectively do one thing at a time. The Palm OS for Palm handheld computers is a good example of a modern single-user, single-task operating system.
- **Single-user, multi-tasking** - This is the type of operating system most people use on their desktop and laptop computers today. Microsoft's Windows and Apple's Mac OS platforms are both examples of operating systems that will let a single user have several programs in operation at the same time. For example, it's entirely possible for a Windows user to be writing a note in a word processor while downloading a file from the Internet while printing the text of an e-mail message.
- **Multi-user** - A multi-user operating system allows many different users to take advantage of the computer's resources simultaneously. The operating system must make sure that the requirements of the various users are balanced, and that each of the programs they are using has sufficient and separate resources so that a problem with one user doesn't affect the entire community of users. Unix, VMS and mainframe operating systems, such as *MVS*, are examples of multi-user operating systems.

Q. No. 38. What is language Translator? Define its types?

Language translator

Computer can only understand programs in machine language but we write program in high level language. Therefore, there must be a language translator which convert high level language into computer (machine) language.

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Types of translator.

There are three types of translator

1. **Assembler**
2. **Compiler**
3. **Interpreter**

Assembler

An assembler is program that convert assembly language program into machine language. Computer code does understand only machine language, assembly languages facilitate

programmers to write programs easily but still needs to convert the assembly language program into machine understandable form.

Compiler

Compiler is program that translates the high-level language program into machine language. Each high level language has its own compiler. Compiler translates the whole program at a time.

Interpreter

Interpreter is a language translator program, which converts high level program into machine language. It translates one instruction at a time. Interpreter does not make any object file and it translate the program every time your will execute it.

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Chapter # 2. 1st Year
Information Networks

Q. No. 1. What is Computer Networks?

Two or more computers and peripheral equipment (e.g., printers) that are connected with one another for the purpose of exchanging data electronically. Two basic network types are local area networks (LANs) and wide-area networks. Wide-area networks connect computers and smaller networks to larger networks over greater geographical areas, including different continents. Communications may occur over cables, fiber optics, or satellites, but most computer users access the network with a modem, using telephone lines.



Q. No. 2. ...uter Networks?

- **File sharing** - Network file sharing between computers gives you more flexibility than using floppy drives or Zip drives. Not only can you share photos, music files, and documents, you can also use a home network to save copies of all of your important data on a different computer. *Backups* are one of the most critical yet overlooked tasks in home networking.

- **Printer / peripheral sharing** - Once a home network is in place, it's easy to then set up all of the computers to share a single printer. No longer will you need to bounce from one system or another just to print out an email message. Other computer peripherals can be shared similarly such as network scanners, Web cams, and CD burners.

- **Internet connection sharing** - Using a home network, multiple family members can access the Internet simultaneously without having to pay an ISP for multiple accounts. You will notice the Internet connection slows down when several people share it, but broadband Internet can handle the extra load with little trouble. Sharing dial-up Internet connections works, too. Painfully slow sometimes, you will still appreciate having shared dial-up on those occasions you really need it.

- **Multi-player games** - Many popular home computer games support *LAN mode* where friends and family can play together, if they have their computers networked.

- **Internet telephone service** - services allow you to make and receive phone calls through your home network across the Internet, saving you money.

- **Home entertainment** - Newer home entertainment products such as digital video recorders (DVRs) and video game consoles now support either wired or wireless home

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networking. Having these products integrated into your network enables online Internet gaming, video sharing and other advanced features.

Q. No. 3. How many types of Network?

Types of Networks

There are several types of computer networks.

LAN (Local Area Network)
WAN (Wide Area Network)
GAN (Global Area Network)

LAN (Local Area Network)

LAN (Local Area Network) supplies networking capability to a group of computers in close proximity to each other such as in an office building, a school, or a home. A LAN is useful for sharing resources like files, printers, games or other applications.

Q. No. 4. What is LAN Topology?

Topology of LAN

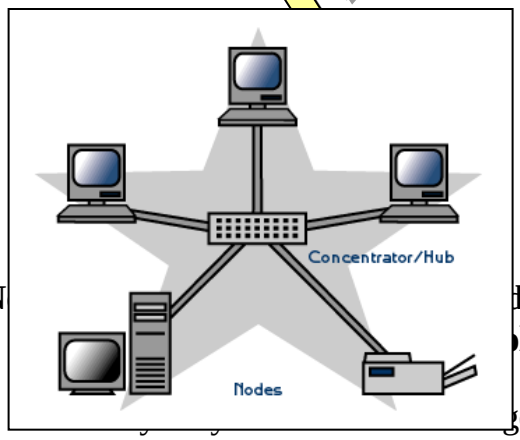
A computer network LAN topology is the physical communication scheme used by connected devices. Or the physical layout of hardware components called Topology. These illustrate the common computer network topologies including bus, ring and star topology diagrams.

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Q. No. 5. Define Star Topology ?

Star topology is one of the most common network setups where each of the devices and computers on a network connect to a central hub. A major disadvantage of this network topology is that if the central hub fails, all computers connected to that hub would be disconnected. Below is a visual example of a simple computer setup on a network using the star topology.

Data on a star network passes through the hub, switch, or concentrator before continuing to its destination. The hub, switch, or concentrator manages and controls all functions of the network. It also acts as a repeater for the data flow. This configuration is common with twisted pair cable; however, it can also be used with coaxial cable or fiber optic cable.



Q. N **d Disadvantages of Star Topology?**

The **ology are:**

- ge star network topology as it is the simplest of the lot when it comes to functionality.

- It is easy to troubleshoot this network type as all computers are dependent on the central hub which invariably means that any problem which leaves the network inoperable can be traced to the central hub.
- In star topology, the data packets don't have to make their way through various nodes which makes sure that the data transfer is fast.
- At the same time, the fact that the data packets only make it through three different points ensures that the data is safe.
- As the nodes are not connected to each other, any problem in one node doesn't hamper the performance of other nodes in the network.
- Adding new machines or replacing the old ones is a lot easy in this network topology, as disruption of the entire network is not required to facilitate the same.

The disadvantages of star network topology are:

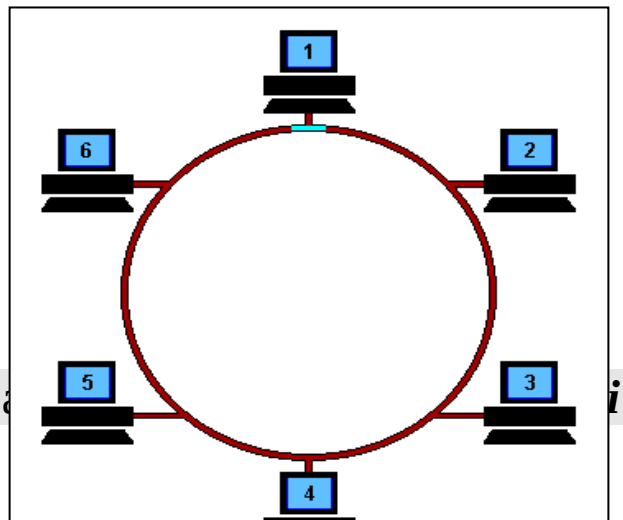
- The foremost problem with star network topology is the fact that it is highly dependent on the functioning of central hub.
- The size of the network is dependent on how many connections can be made to the hub.
- This network type requires more cable as compared to linear bus topology which means the expenses incurred would be relatively high.
- The performance of the entire network is directly dependent on the performance of the hub. If the server is slow, it will cause the entire network to slow down.
- If one of the numerous nodes utilizes a significant portion of the central hub's processing capability, it will reflect on the performance of other nodes.

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Q. No. 7. Define Ring Topology?

In the ring topology is a network within which all computer are connected like in a circle. The ring topology data transmission is depending on a token.

Ring topologies are used on token ring networks. Each device processes and retransmits the signal, so it is capable of supporting many devices in a somewhat slow but very orderly fashion. A token, or small data packet, is continuously passed around the network. When a device needs to transmit, it reserves the token for the next trip around, then attaches its data packet to it. The receiving device sends back the packet with an acknowledgment of receipt, then the sending device puts the token back out on the network. Most token ring networks have the physical cabling of a star topology and the logical function of a ring through use of multi access units (MAU). In a ring topology, the network signal is passed through each network card of each device and passed on to the next device.



Q. No. 8. Discuss the Advantages and Disadvantages of Ring Topology/

Advantages of Ring Topology

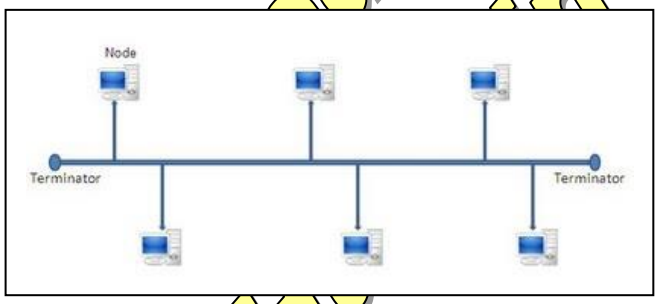
1) This type of network topology is very organized. Each node gets to send the data when it receives an empty token. This helps to reduces chances of collision. Also in ring topology all the traffic flows in only one direction at very high speed.
2) Even when the load on the network increases, its performance is better than that of Bus topology.
3) There is no need for network server to control the connectivity between workstations.
4) Additional components do not affect the performance of network.
5) Each computer has equal access to resources.

Disadvantages of Ring Topology

1) Each packet of data must pass through all the computers between source and destination. This makes it slower than Star topology.
2) If one workstation or port goes down, the entire network gets affected.
3) Network is highly dependent on the wire which connects different components.
4) MAU's and network cards are expensive as compared to Ethernet cards and hubs.

Q. No. 9. Define BUS Topology?

This network topology has computers connected to a strand of network cabling that is connected to network repeaters at one end and terminated at the other. If you break part of the cable or remove the terminator, all machines on that segment lose communication with the network.



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Q. No. 10. What are the Advantages and disadvantages of Bus Topology?

Advantages (benefits) of Linear Bus Topology

1) It is easy to set-up and extend bus network.
2) Cable length required for this topology is the least compared to other networks.
3) Bus topology costs very less.
4) Linear Bus network is mostly used in small networks. Good for LAN.

Disadvantages (Drawbacks) of Linear Bus Topology

1) There is a limit on central cable length and number of nodes that can be connected.
2) Dependency on central cable in this topology has its disadvantages. If the main cable (i.e. bus) encounters some problem, whole network breaks down.
3) Proper termination is required to dump signals. Use of terminators is must.

4) It is difficult to detect and troubleshoot fault at individual station.
5) Maintenance costs can go higher with time.
6) Efficiency of Bus network reduces as the number of devices connected to it increases.
7) It is not suitable for networks with heavy traffic.
8) Security is very low because all the computers receive the sent signal from the source.

Q. No. 11. What are the Cabling system as components of a LAN?

Connection or Cabling System

Cable is the medium through which information usually moves from one network device to another. There are several types of cable which are commonly used with LANs.

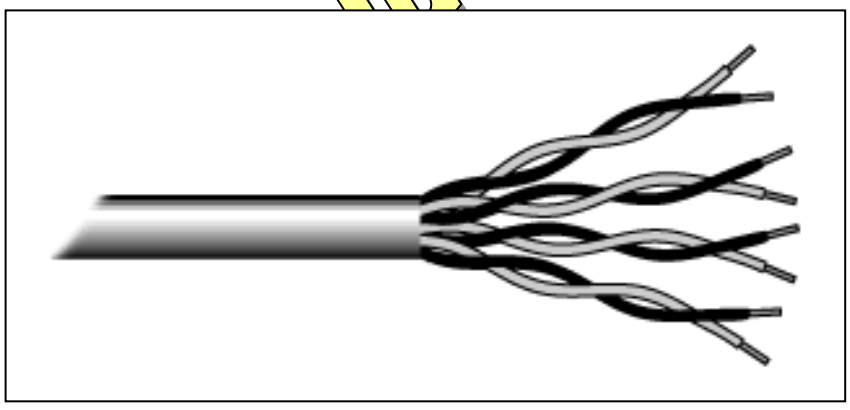
The following sections discuss the types of cables used in networks and other related topics.

- **Unshielded Twisted Pair (UTP) Cable**
- **Shielded Twisted Pair (STP) Cable**
- **Coaxial Cable**
- **Fiber Optic Cable**

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Unshielded Twisted pair cabling

Twisted pair cabling comes in two varieties: shielded and unshielded. Unshielded twisted pair (UTP) is the most popular and is generally the best option for school networks.



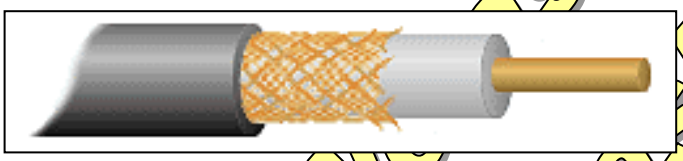
The quality of UTP may vary from telephone-grade wire to extremely high-speed cable. The cable has four pairs of wires inside the jacket. Each pair is twisted with a different number of twists per inch to help eliminate interference from adjacent pairs and other electrical devices.

Shielded Twisted Pair (STP) Cabling

Although UTP cable is the least expensive cable, it may be susceptible to radio and electrical frequency interference (it should not be too close to electric motors, fluorescent lights, etc.). If you must place cable in environments with lots of potential interference, or if you must place cable in extremely sensitive environments that may be susceptible to the electrical current in the UTP, shielded twisted pair may be the solution. Shielded cables can also help to extend the maximum distance of the cables.

Coaxial Cable

Coaxial cabling has a single copper conductor at its center. A plastic layer provides insulation between the center conductor and a braided metal shield. The metal shield helps to block any outside interference from fluorescent lights, motors, and other computers.

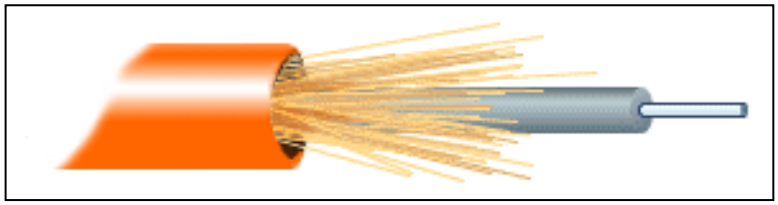


Although coaxial cabling is difficult to install, it is highly resistant to signal interference. In addition, it can support greater cable lengths between network devices than twisted pair cable. The two types of coaxial cabling are thick coaxial and thin coaxial.

Thin coaxial cable is also referred to as thinnest. 10Base2 refers to the specifications for thin coaxial cable carrying Ethernet signals. The 2 refers to the approximate maximum segment length being 200 meters. In actual fact the maximum segment length is 185 meters. Thin coaxial cable has been popular in school networks, especially linear bus networks.

Fiber Optic Cable

Fiber optic cabling consists of a center glass core surrounded by several layers of protective materials (See fig. 5). It transmits light rather than electronic signals eliminating the problem of electrical interference. This makes it ideal for certain environments that contain a large amount of electrical interference. It has also made it the standard for connecting networks between buildings, due to its immunity to the effects of moisture and lighting.

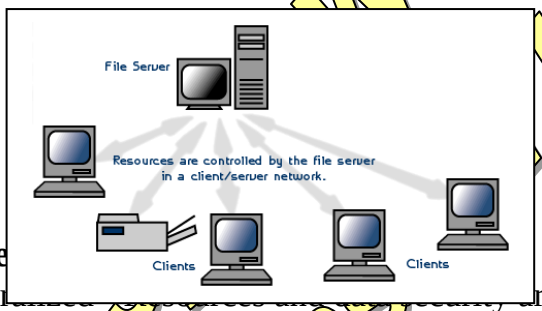


Q. No. 12. Define Wireless LAN?

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More and more networks are operating without cables, in the wireless mode. Wireless LANs use high frequency radio signals, infrared light beams, or lasers to communicate between the workstations, servers, or hubs. Each workstation and file server on a wireless network has some sort of transceiver/antenna to send and receive the data. Information is relayed between transceivers as if they were physically connected. For longer distance, wireless communications can also take place through cellular telephone technology, microwave transmission

Q. No. 13. What is LAN Models? Also discuss the advantages and disadvantages?
There are two types of LAN Models.
Client / Server
Client/server network operating systems allow the network to centralize functions and applications in one or more dedicated file servers. The file servers become the heart of the system, providing access to resources and providing security. Individual workstations (clients) have access to the resources available on the file servers. The network operating system provides the mechanism to integrate all the components of the network and allow multiple users to simultaneously share the same resources irrespective of physical location. UNIX/Linux and the Microsoft family of Windows Servers are examples of client/server network operating systems.



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Advantage
- Centr re controlled through the server.
- Scalability - Any or all elements can be replaced individually as needs increase.
- Flexibility - New technology can be easily integrated into system.
- Interoperability - All components (client/network/server) work together.
- Accessibility - Server can be accessed remotely and across multiple platforms.

Disadvantages of a client/server network:
- Expense - Requires initial investment in dedicated server.
- Maintenance - Large networks will require a staff to ensure efficient operation.
- Dependence - When server goes down, operations will cease across the network.

Peer-to-Peer

Peer-to-peer network operating systems allow users to share resources and files located on their computers and to access shared resources found on other computers. However, they do not have a file server or a centralized management source. In a peer-to-peer network, all computers are considered equal; they all have the same abilities to use the resources available on the network. Peer-to-peer networks are designed primarily for small to medium local area

networks. Nearly all modern desktop operating systems, such as Macintosh OSX, Linux, and Windows, can function as peer-to-peer network operating systems.

Advantages of a peer-to-peer network:

- Less initial expense - No need for a dedicated server.
- Setup - An operating system (such as Windows XP) already in place may only need to be reconfigured for peer-to-peer operations.

Disadvantages of a peer-to-peer network:

- Decentralized - No central repository for files and applications.
- Security - Does not provide the security available on a client/server network.

Q. No. 14. What are the components of a LAN?

1) Connection or Cabling System

LAN used two types of connection or cabling system

1. **Guided Medium**
2. **Un-guided Medium**

Guided Medium

1. Twisted pair cable
2. Coaxial cable
3. Fiber optic cable

Un-guided medium

1. Wireless connection
2. Radio-wave transmission
3. Bluetooth
4. WiFi

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2) Microcomputer with Network Interface Cards (NIC)

Two or more microcomputer are required along with network interface cards. Network interface card enables the computer to send and received messages over a cable network

3) Network Operating System

Unlike operating systems, such as Windows, that are designed for single users to control one computer, network operating systems (NOS) coordinate the activities of multiple computers across a network. The network operating system acts as a director to keep the network running smoothly.

The examples of most popular network operating system are Microsoft Windows, UNIX/ Linux and Macintosh OS X which support the peer-to-peer and client/server network operating systems.

4) Other Shared Devices

Printer, fax, scanners, storage and other peripherals may be added to the network as necessary and shared by all users

5) Router, Bridges and Gateways

Router

Routers are the traffic directors of the global internet. All routers maintain complex routing tables which allow them to determine appropriate paths for packets destined for any address. Routers communicate with each other, and forward network packets out of or into a network

Bridges

A bridge is a device that allows you to segment a large network into two smaller, more efficient networks. If you are adding to an older wiring scheme and want the new network to be up-to-date, a bridge can connect the two.

Gateways

A gateways is an interface that enables dissimilar network to communicate, such as a LAN with a WAN or two LAN based on different topology or network operating systems

Q. No. 15. What is WAN (Wide Area Network)?

A wide area network (WAN) is a geographically dispersed telecommunications network. The term distinguishes a broader telecommunication structure from a local area network (LAN). A wide area network may be privately owned or rented, but the term usually connotes the inclusion of public (shared user) networks. An intermediate form of network in terms of geography is a metropolitan area network (MAN).

Q. No. 16. What is Global Area Network (GAN)?

Global Area Network or Internet a (worldwide collection of interconnected networks)is a collection of network.

A network of networks. That communicate with each other by using same standard (Protocols) of communication. The computer that make up these networks are of different types.

Q. No. 17. What is ARPANET?

The **Advanced Research Projects Agency Network** (**ARPANET**), was the world's first operational packet switching network and the core network of a set that came to compose the global Internet. The network was funded by the Defense Advanced Research Projects Agency (DARPA) of the United States Department of Defense for use by its projects at universities and research laboratories in the US. The packet switching of the ARPANET was based on designs by Lawrence Roberts

Q. No. 18. What is Internet?

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The **Internet** is a global system of interconnected computer networks that use the standard Internet protocol suite (TCP/IP) to serve billions of users worldwide. It is a network of networks that consists of millions of private, public, academic, business, and government networks, of local to global scope, that are linked by a broad array of electronic, wireless and optical networking technologies. The Internet carries a vast range of information resources and services, such as the inter-linked hypertext documents of the World Wide Web (WWW) and the infrastructure to support email.

Q. No. 19. What are the advantages of Internet?

Advantages of internet

There many advantages to using the internet such as:

- **Email.**
  Email is now an essential communication tools in business. It is also excellent for keeping in touch with family and friends. The advantages to email is that it is free (no charge per use) when compared to telephone, fax and postal services.
- **Information.**
  There is a huge amount of information available on the internet for just about every subject known to man, ranging from government law and services, trade fairs and conferences, market information, new ideas and technical support.
- **Services.**
  Many services are now provided on the internet such as online banking, job seeking and applications, and hotel reservations. Often these services are not available off-line or cost more.
- **Buy or sell products.**
  The internet is a very effective way to buy and sell products all over the world.
- **Communities.**
  Communities of all types have sprung up on the internet. Its a great way to meet up with people of similar interest and discuss common issues.

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Chapter # 3. 1st Year
Data Communications

Q. No. 1. Define Data Communication?
I would define data communication as the transfer of information between two points, either via an analogue (sine wave) electrical signal. or digital (binary) signal via electrical pulses or optically via light pulses

Data and information from one computer system can be transmitted to other systems across geographical areas. Thus data transmission is the movement of information using some standard methods. These methods include electrical signals carried along a conductor, optical signals along an optical fibers and electromagnetic areas.

Q. No. 2. What are the basic elements of Data Communication?

Basic Components



Source: **The transmitter of data. Examples are:**
- Terminals
- Computers
- Mainframes

Medium: **The communications stream through which the data is being transmitted. Examples are:**
- Cabling
- Microwave
- Fiber optics
- Radio Frequencies (RF)
- Infrared Wireless

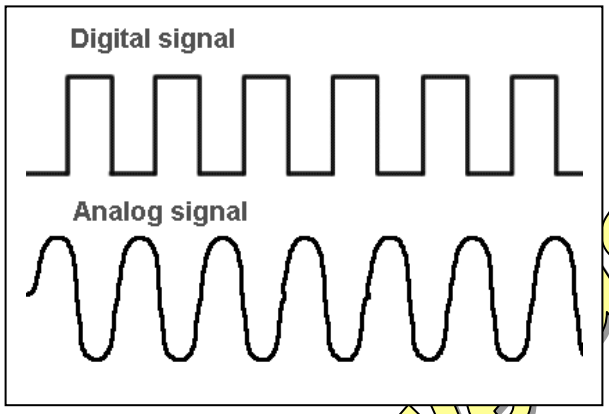
Receiver: **The receiver of the data transmitted. Examples are:**
- Printers
- Terminals
- Mainframes
- Computers

Q. No. 3. What is communication signal?
Data is transmitted from one point to another point by means of electrical signals

that may be in digital and analog form

Q. No. 4. Define the Digital and Analog signals?
Digital and Analog Signals

Digital Signal
A digital signal uses of/off or present / absent electronic pulses in discrete rather than a continues waves form.

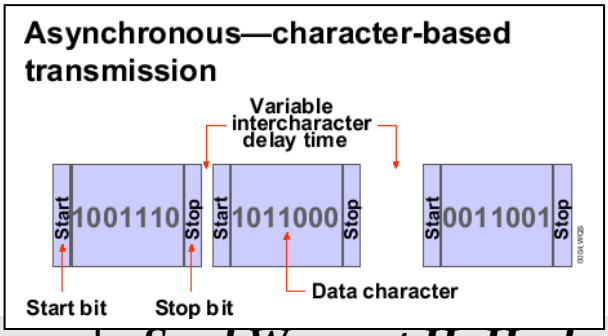


Analog Signal
An **analog** or **analogue signal** is any continuous signal for which the time varying feature (variable) of the signal is a representation of some other time varying quantity, i.e., analogous to another time varying signal. It differs from a digital signal in terms of small fluctuations in the signal which are meaningful. Analog is usually thought of in an electrical context

Q. No. 5. What are the Transmission Modes?
There are two types of Transmission Modes.
Asynchronous Transmission
Synchronous Transmission

Asynchronous transmission

uses start and stop bits to signify the beginning bit ASCII character would actually be transmitted using 10 bits. For example, "0100 0001" would become "**1** 0100 0001 **0**". The extra one (or zero, depending on parity bit) at the start and end of the transmission tells the receiver first that a character is coming and secondly that the character has ended. This method of transmission is used when data are sent intermittently as opposed to in a solid stream. In the previous example the start and stop bits are in bold. The start and stop bits must be of opposite polarity. This allows the receiver to recognize when the second packet of information is being sent.



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Synchronous transmission

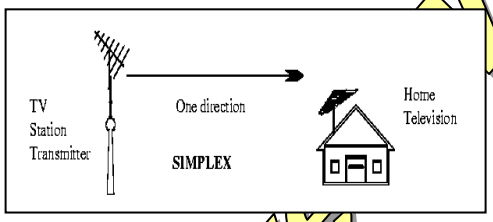
uses no start and stop bits, but instead synchronizes transmission speeds at both the receiving and sending end of the transmission using clock signal(s) built into each component. A continual stream of data is then sent between the two nodes. Due to there being no start and stop bits the data transfer rate is quicker although more errors will occur, as the clocks will eventually get out of sync, and the receiving device would have the wrong time that had been agreed in the protocol for sending/receiving data, so some bytes could become corrupted (by losing bits). Ways to get around this problem include re-synchronization of the clocks and use of check digits to ensure the byte is correctly interpreted and received

Q. No. 6. What are the Direction of Transmission?

There are three ways for transmitting data from one point to another

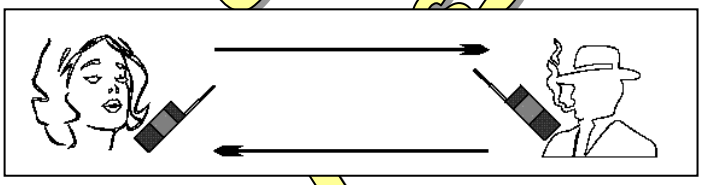
1. **Simplex** :

n simplex mode the communication can take place in one direction. The receiver receives the signal from the transmitting device. In this mode the flow of information is Uni-directional. Hence it is rarely used for data communication.



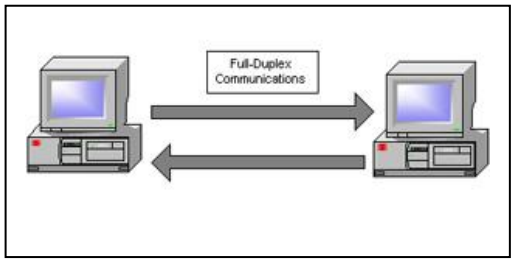
2. **Half-duplex** :

In half-duplex mode the communication channel is used in both directions, but only in one direction at a time. Thus a half-duplex line can alternately send and receive data.



3. **Full-duplex** :

In full duplex the communication channel is used in both directions at the same time. Use of full-duplex line improves the efficiency as the line turnaround time required in half-duplex arrangement is eliminated. Example of this mode of transmission is the telephone line.



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Q. No. 7. Define Bandwidth and its types?
Bandwidth refer to how much data can send through a network. Computer may be connected by telephone wires, cable or through radio waves or microwaves.
Bandwidth rate is usually measured in bites per second (bps) and is often expressed in units of thousands(k) or millions (M0 for example:
1 kbps = 1000 bps
1 mbps = 1000000 bps

Types of Bandwidth

Narrowband
These media communication data with relatively slow speed. Example of narrow band is telegraph lines.

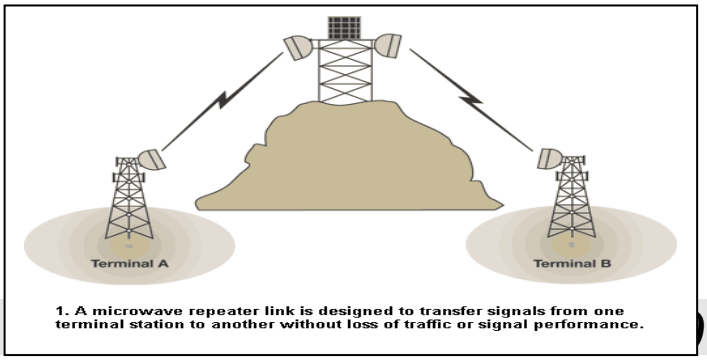
Voice band
These media are faster than narrowband. Most telephone lines which are used to carry microcomputer transmissions are voice band.

Broadband
It is also called wideband. This media transmit large volumes of data at high speed via microwave, satellite, coaxial cable and fiber optic cable.

Q. No. 8. Define communication Media?
See also Q. No. 11 of chapter no 2.

Q. No. 9 Define Microwave transmission?
The term microwave refers to electromagnetic energy having a frequency higher than 1 gigahertz (billions of cycles per second), corresponding to wavelength shorter than 30 centimeters.
Microwave signals propagate in straight lines and are affected very little by the troposphere. They are not refracted or reflected by ionized regions in the upper atmosphere. Microwave
Microwave transmission also requires line of sight in order to work properly. In order to allow two way communications two frequencies are used. However, this does not mean that there has to be two antennas because the frequencies can be dealt with by one antenna at both ends.

The distance covered by microwave signals is based upon the height of the antenna. In order to increase this coverage each antenna has a built-in repeater that regenerates the signal before passing it on to the next antenna in line. The placement of the antenna to do this is approximately 25 miles.

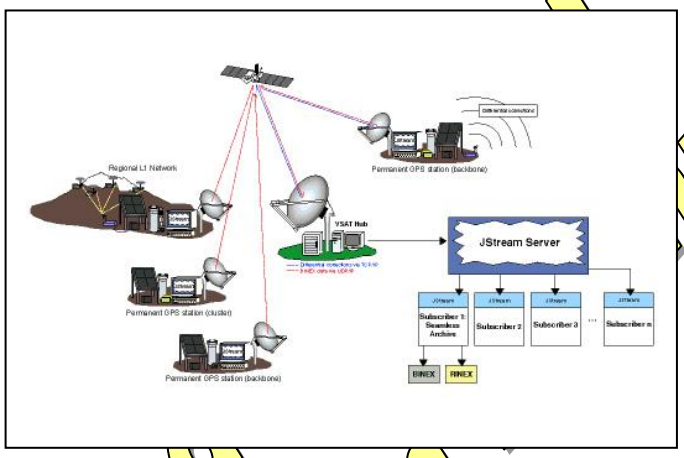


1. A microwave repeater link is designed to transfer signals from one terminal station to another without loss of traffic or signal performance.

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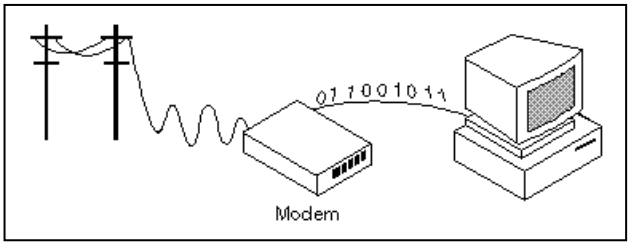
Q. No. 10 . Define Satellite Transmission?

Satellite transmission provide communication likes between various points on earth. Satellite communication play a vital role in the global telecommunication system. Approximately 2000 artificial satellites orbiting earth relay analog and digital signals carrying voice, video, and data to a and from one or many location worldwide.
Satellite communication has two main components. One of the ground segment and second is space segment.
A typical satellite link involves the transmission or up-linking of a signal from an earth station to a satellite.



Q. No. 11. Define communication Hardware?
Each computer on the network needs hardware component to control the flow of data. Some of the common data communications hardware are:
Modem
Ethernet
Fast Ethernet
Token Ring

Q. No. 12. What is Modem?
A modem (Modulator / DEModulator) connects a terminal/computer to the Voice Channel (dial-up line). Digital modems developed from the need to transmit data for North American air defense during the 1950s. Modems were used to communicate data over the public switched telephone network.
The modem is responsible for converting digital data signal used by computer into an analog signal used on phone lines and then converting it back once received on the other end.



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Q. No. 13. Define types of Modem
There are two types of Modem
External Modem
Internal Modems

- **External Modem:** This is a modem separated from the system unit in the computer case. It is connected to the serial port of the computer by means of a cable. It is connected to the telephone wall jack by another cable.
- **Internal Modem:** An internal modem is a circuit board (a modem card) that can be added to the system unit of the computer. It takes one of the expansion slots.

Q. No. 14. What is Ethernet?

Ethernet

Ethernet is a network standard of communication using either coaxial or twisted pair cable. IEEE is Institute of Electrical and Electronic Engineers developed by International Standard Organization (ISO) in 1985. IEEE developed LAN standards, which is commonly known as IEEE 802 standards. These are set of standards which are widely adopted by IT sector through out the world. Most commonly known and popular standards of IEEE are IEEE 802.3 which is known as Ethernet. Ethernet is LAN technology used world wide on very large scale, almost every office, school and even homes have deployed Ethernet technology

Fast Ethernet

Fast Ethernet is a local area network (LAN) transmission standard that provides a data rate of 100 megabits per second (referred to as "100BASE-T"). Workstations with existing 10 megabit per second (10BASE-T) Ethernet card can be connected to a Fast Ethernet network. (The 100 megabits per second is a shared data rate; input to each workstation is constrained by the 10 Mbps card.)

Q. No. 15. What is Token Ring?
See Answer No. 7 from the chapter No. 2.

Q. No. Define Communication Protocols?
A protocol is a set of rules that governs the communications between computers on a network. In order for two computers to talk to each other, they must be speaking the same language. Many different types of network protocols and standards are required to ensure that your computer (no matter which operating system, network card, or application you are using) can communicate with another computer located on the next desk or half-way around the world

- **TCP/IP** – Transmission Control Protocol / Internet Protocol is a protocol suit used to control data exchanged between two ends these ends can be computer or other device.
- **DNS** - Domain Name System - translates network address (such as IP addresses) into terms understood by humans (such as Domain Names) and vice-versa
- **DHCP** - Dynamic Host Configuration Protocol - can automatically assign Internet addresses to computers and users
- **FTP** - File Transfer Protocol - a protocol that is used to transfer and manipulate files on the Internet

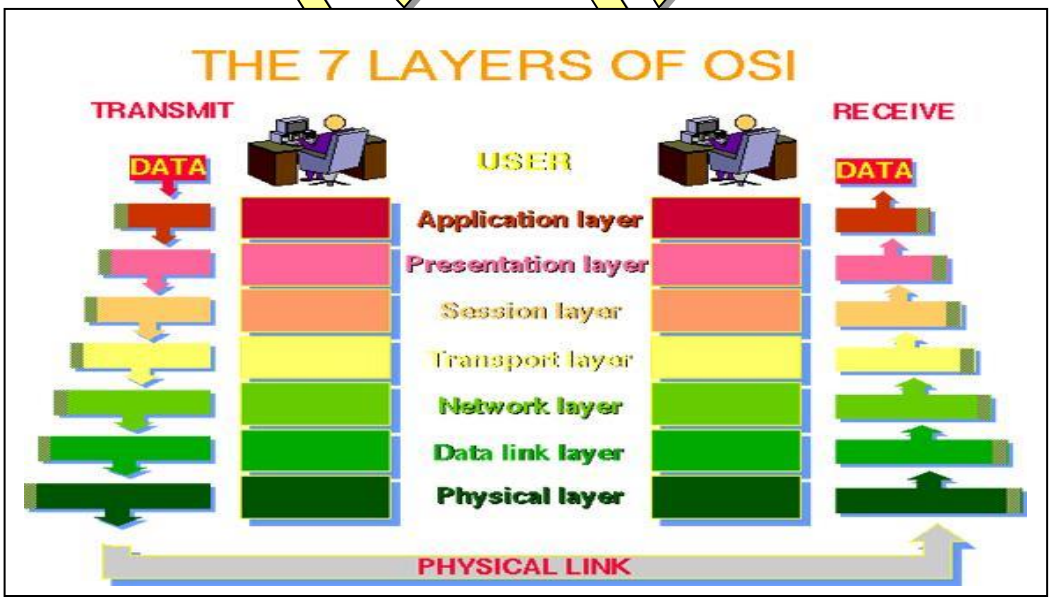
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- **HTTP** - HyperText Transfer Protocol - An Internet-based protocol for sending and receiving webpages
- **IMAP** - Internet Message Access Protocol - A protocol for e-mail messages on the Internet
- **IRC** - Internet Relay Chat - a protocol used for Internet chat and other communications
- **POP3** - Post Office protocol Version 3 - a protocol used by e-mail clients to retrieve messages from remote servers
- **SMTP** - Simple Mail Transfer Protocol - A protocol for e-mail messages on the Internet

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Q. No. 16. Define OSI Model?

Open Systems Interconnection (OSI) is a standard reference model for communication between two end users in a network. The model is used in developing products and understanding networks.



7) Application Layer : The application layer provider different services to the application. Example of services provided by this layer are file transfer, electronic messaging e-mail, virtual terminal access and network management.

6) Presentation Layer : The Presentation layer is responsible for protocol conversion, date encryption/decryption, Expanding graphics command and the date compression. This layer makes the communications between two host possible.

5) Session Layer : This layer is responsible for establishing the process-to-process communication between the host in the network. This layer is responsible for establishing and ending the sessions across the network. The interactive login is an example of services provided by this layer in which the connective are re-connected in care of any interruption.

4) Transport Layer : This layer is responsible for end-to-end delivers of messages between the networked hosts. It first divides the streams of data into chunks or packets before transmission and then the receiving computer re-assembles the packets. It also guarantee error free data delivery without loss or duplications.

3) Network Layer : This layer is responsible for translating the logical network address and names into their physical address (MAC address). This layer is also responsible for addressing, determining routes for sending and managing network problems such as packet switching, data congestion and routines.

2) Data Link Layer : Data link layer is responsible for controlling the error between adjacent nodes and transfer the frames to other computer via physical layer. Data link layer is used by hubs and switches for their operation.

1) Physical Layer : Physical Layer is responsible for transmitting row bit stream over the physical cable. The physical layer defines the hardware items such as cables, cards, voltages etc.

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Chapter # 04
Application and Uses of Computers

Q. No. 1. What is the use of computer in Education?

Use of computer in Education

With the advent of the Internet age, all aspects of society have been influenced, including education. Computers and the Internet are used in all levels of education, excluding perhaps preschool and grade school.

1. Higher Education

- All universities and colleges in the United States are furnished with computer labs. Because many professors require their students to turn in typed documents, it is more convenient to have computer labs on campus for student use.

2. Faculty Usage

- Teachers at all levels use computers to administer assignments, keep track of grades or offer online instruction. Many teachers use online resources in their daily lessons.

3. Online Education

- There is an entire "sub-industry" of education called "online education." Online education is done primarily through the usage of computers and the Internet where the student does the class work from home.

4. Self-Learning

- In addition to books, video and other materials, self-directed learners often use computers to continue their education outside of, or in addition to, traditional facilities.

5. Social Networking

- Popular networking sites like My Space and Facebook can also be used for educational purposes, as some instructors integrate aspects of those sites into their lessons.

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Q. No. 2. Write some of the popular productivity tools of computer?

Computer productivity is the art of working efficiently with your computer by improving your interactions with it. This is done primarily by using your mouse and keyboard efficiently, but it also involves cleverly customizing your computer's interface. It's true that you can always make your computer time more efficient by getting a faster computer
Productivity tools can be software that help employer increase their business productivity

Word Processing

Using a computer to create, edit, and print documents. Of all computer applications, word processing is the most common. To perform word processing, you need a computer, a special program called a *word processor*, and a printer. A word processor enables you to create a document, store it electronically on a disk, display it on a screen, modify it by entering commands and characters from the keyboard, and print it on a printer

Desktop Publishing
Use of a personal computer to perform publishing tasks. DTP allows an individual to combine text, numerical data, and graphic elements in a document that can be output on a printer or a phototypesetter. A typical DTP system includes a personal computer, a high-resolution printer, and input devices such as an optical scanner. Text and graphic elements are commonly created or manipulated with several separate software programs and then combined with a page-makeup program. Powerful DTP programs offer full-featured graphics capabilities

Database
A database can be thought of as a set of logically related files organized to facilitate access by one or more applications programs and to minimize data redundancy. This concept does not imply that all data relating to a company's business should be contained on a single database, but simply that all records in a database should be related and that redundant data should be minimized.

Graphic
Computer graphics are graphics created using computers and, more generally, the representation and manipulation of image data by a computer with help from specialized software and hardware.

Communication
Computers and communication networks make up an infrastructures of our information based culture and society. These networks provide socio-economic benefits. At the same time they help improve many critical operations in business services (ie: in health care, finance and transportation).

Q. No. 3. Define the use of computer in Medical field?
There are various uses of computer in medical field as it plays a vital role in every medical office. It contributes a lot to the hospital administration where all the administrative processes are computerized. For example, staff records, recording incoming and outgoing time of the staff, the holiday records of the staff, etc. It is difficult to keep track of such things manually but with the help of computers, the task is simplified and is less time-consuming. In most of the offices, accounting is also computerized that helps you keep the daily record of the financial transactions of the office

Q. No. 4. Define the use of computer in different fields?
Energy
Companies are using computer to locate the oil, petrol, natural gas etc, meter readers using handy computer to record the meter reading.

Banks
Computers are used in banks for a variety of reasons. They help bank personnel operate more efficiently and effectively. Computers are used to track certain transactions and they help process other customer information as well. Without computers, it would be very hard for a bank to offer good customer service day in and day out. Computers help a bank save time and money, and can be used as an aid to generate profits.
Agriculture

1. Scientists have created a new, affordable technology that allows farmers to detect whether their crops are contaminated with a fungal toxin that causes health problems.

2. Farmers use a milking machine to milk a large number of cows faster

3. Scientists are developing an agricultural robot for identifying and eliminating weeds. While this might seem like a relatively easy task, it actually requires a lot of machine intelligence to pick out the weeds among the crops. The robot is still in the early stages of development, but the researchers hope that it will ultimately lead to a reduction in the amount of herbicides used by farmers and therefore cut costs

Transport

The use of computers in transportation has exploded over the past decade. Vehicles and transportation infrastructure use all types of electronic computational components: application specific standard processors (ASSPs), systems-on-chips (SoCs), single board computers (SBCs), and fully integrated large-scale systems. The choice is dependent on many variables but is primarily driven by size, weight and power.

Law Enforcement

Computer technology has been both a bane and a benefit to law enforcement. Computer technology has created an entire new realm of criminal activity for law enforcement to deal with--in the form of hackers and Internet predators. Those same technical advancements have increased the power and range of law enforcement to capture criminals. Most police forces, local sheriff departments and government agencies are computerized and employ some form of technology in their everyday crime-fighting efforts.

Home

New research has found that children are learning to use their home computers in a totally different way from the way they use them at school, which raises questions about current educational thinking on ICT teaching at school, as well as future policies aimed at linking homes and schools through new technology.

Entertainment

People in the entertainment industries have found a wide range of uses for computers. Musicians use computers to create an amazing range of instruments and sounds simply by playing a keyboard. The Musical Instrument Digital Interface (MIDI) is a system that synchronizes hardware and software that produce electronic tones. Sound is a pattern of waves, and computers can bend and alter those waves into musical shapes.

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Chapter # 5
Computer Architecture

Q. No. 1. What is meant by Computer Architecture?
Computer architecture is the practical art of selecting and interconnecting hardware components to create computers that meet functional, performance and cost goals and the formal modeling of those systems.
The Computer architecture requirements and basic design for the various parts of a computer. It is usually most concerned with how the central processing unit (CPU) acts and how it accesses computer memory. Some currently (2011) fashionable computer architectures include cluster computing and Non-Uniform Memory Access.

Q. No. 2. Define computer Organization?
Computer organization helps optimize performance-based products. For example, software engineers need to know the processing ability of processors. They may need to optimize software in order to gain the most performance at the least expense. This can require quite detailed analysis of the computer organization. For example, in a multimedia decoder, the designers might need to arrange for most data to be processed in the fastest data path and the various components are assumed to be in place and task is to investigate the organizational structure to verify the computer parts operates.

Q. No. 3. What are the main components of Digital Computer?
Main components of Digital Computer are following :
Input Unit
Output Unit
Central Processing Unit
Memory Unit

Input Unit
This is the process of entering data and programs in to the computer system. You should know that computer is an electronic machine like any other machine, which takes as input raw data and performs some processing giving out processed data. Therefore, the input unit takes data from us to the computer in an organized manner for processing.

Output Unit
Any peripheral that receives or displays output from a computer. In the picture to the right, is an inkjet printer and a good example of an **output device**

Central Processing Unit

The computer is an information processing machine and the processor (CPU) is that part of the computer
where information is processed. More specifically, the processor executes the program.

CPU divided into four main parts
Control Unit (CU)
Arithmetic Logic Unit (ALU)

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Registers
Memory Unit (MU)

The main feature of the computer, the feature that makes it so different from other machines, is the *Internal control*. The computer is controlled by a *program*, which specifies the operations to be performed. The program is located in memory, outside the processor, and the processor executes the program by reading (Fetching) it from memory, instruction by instruction, and executing each instruction. This task is performed mainly by the control unit (CU), which uses the ALU and the registers as tools. The ALU is that part of the processor where operations are performed on numbers. The registers, as mentioned earlier, are fast, Temporary storage.

Q. No. 4 .Define Control Unit (CU)?
The Control Unit can be thought of as the brain of the CPU itself. It controls based on the instructions it decodes, how other parts of the CPU and in turn, rest of the computer systems should work in order that the instruction gets executed in a correct manner. There are two types of control units, the first type is called hardwired control unit. Hardwired control units are constructed using digital circuits and once formed cannot be changed. The other type of control unit is micro programmed control unit. A micro programmed control unit itself decodes and execute instructions by means of executing micro programs.

Q. No. 5. Define Arithmetic Logic Unit (ALU)
After enter data through the input device it is stored in the primary storage unit. Arithmetic Logical Unit performs the actual processing of data and instruction. The major operations performed by the ALU are addition, subtraction, multiplication, division, logic and comparison. Data is transferred to ALU from storage unit when required. After processing, the output is returned back to storage unit for further processing or getting stored.

Q. No. 6. What is Memory Unit (MU)?

Internal storage areas in the computer. The term *memory* identifies data storage that comes in the form of chips, and the word *storage* is used for memory that exists on tapes or disks. Moreover, the term *memory* is usually used as a shorthand for *physical memory,* which refers to the actual chips capable of holding data.

Types of Memory

There are two types of memory: The primary memory or random access memory (RAM), which is temporary storage. And the secondary memory or read-only memory (ROM), which is permanent storage.

Q. No. 7. What is Registers? Also define types of Registers?
A register is one of a small set of data holding places that are part of a computer processor . A register may hold a computer instruction , a storage address, or any kind of data (such as a bit sequence or individual characters). Some instructions specify registers as part of the instruction. For example, an instruction may specify that the contents of two defined registers be added together and then placed in a specified register. A register must be large enough to

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hold an instruction - for example, in a 32-bit instruction computer, a register must be 32 bits in length

Types of register
Program Counter (PC)
The program counter, PC, is a special-purpose register that is used by the processor to hold the address of the next instruction to be executed.

Instruction Register (IR)
In computing, an instruction register (IR) is the part of a CPU's control unit that stores the instruction currently being executed or decoded

Memory Address Register (MAD)
The Memory Address Register (MAR) is a CPU register that either stores the memory address from which data will be fetched to the CPU or the address to which data will be sent and stored.

Memory Buffer Register (MBR)
Memory Buffer Register (MBR) is the register in a computer's processor, or central processing unit, CPU, that stores the data being transferred to and from the immediate access store. It acts as a buffer allowing the processor and memory units to act independently without being affected by minor differences in operation

Memory Data Register (MDR)
The Memory Data Register (MDR) is the register of a computer's control unit that contains the data to be stored in the computer storage (e.g. RAM), or the data after a fetch from the computer storage. It acts like a buffer and holds anything that is copied from the memory ready for the processor to use it.

Accumulator (ACC)
Accumulator (computing), in a CPU, a processor register for storing intermediate results

Q. No. 8. Define Computer Instruction and components of Computer Instruction?
Instruction set is a step that include information on how to operate, perform, or maintain particular computer software.
An instruction is stored in the computer memory as certain specified number of bit or bits

Q. No. 9. Define the components of computer instruction set/

There are two part of computer instruction set.
Operation Code, (Opcode)
Address or Addresses

Operation Code
Opcode is an instruction contain group of bits that defines some arithmetic or some other operations. It may be subtuction, addition, multiplication and division. The number of bit required for an operation code depends upon the total number of operations to be performed by the computer.

Address
An instruction must specify the operation to be performed along with the address or addresses of the operands or registers where the result of an operation is to be stored.

- An instruction has two components:

Opcode	Operand(s) Address

- The operand field may have the following formats:
  1) zero-address
  2) one-address
  3) two-address
  4) three-address

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Chapter # 6.
Security, Copyright and the Law

Q. No 1. What is meant by computer crime?

Computer crime also called Cyber Crime. Any illegal or criminal activity like fund transfer, hacking of data, and Web side, copyright infringement, software piracy, virus, Blackmail and Extortion that involve a computer and a network called computer crime.

Q. No. 2. What is computer Virus?

A computer virus is a self-coping program that design to spread itself or by email or by inserting copies of itself into other programs or documents. This not a biological virus it is logical virus that can copy or replicated one storage place to another and infect (Damage) the computer and generate different problems.

Q. No. 3. How computer viruses infect the computer?

Viruses can enter computers in many ways. Once a virus has entered a system, it will generally hide until it is unknowingly run by the user. A virus will not act until it has been run or some pre-established condition has been met, such as a specific date. The effects of a virus may not be noticed for some time after it has infected a computer.

Viruses can infect several components of a computer's operating and file system including:

- System Sectors/Boot Records - Viruses can infect the parts of the system that are used to run programs and perform functions such as start up and shut down.
- Files - Viruses can infect program files. These viruses stick to program files such as .com, .exe, .sys, etc. Some viruses hide in the memory of the computer at first, while others simply attack a specific software program, such as Microsoft Word.
- Companion Files - Viruses can create companion files that are a special type of file that adds files that run on the hard disk.
- Macros - Viruses can infect macro or data files.
- Disk Clusters - Viruses can infect files through the disk directory.
- Batch Files - Viruses can use batch files to infect a computer.
- Source Code - Viruses can be in additional code that is added to actual program source code.

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Q. No. 4. Define the different types of viruses?

Viruses are categorized by how they infect computers. Some viruses fall into more that one of these categories.

Types of viruses include:

- **Polymorphic Viruses –**
Polymorphic viruses change characteristics as they infect a computer.
- **Stealth Viruses –**
Stealth viruses actively try to hide themselves from anti-virus and system software.
- **Fast and Slow Infectors –**
Fast and Slow viruses infect a computer in a particular way to try to avoid being detected by anti-virus software.
- **Sparse Infectors –**
Sparse Infectors don't infect very often.
- **Armored Viruses –**
Armored viruses are programmed to make eradication difficult.
- **Multipartite Viruses –**
Multipartite Viruses are viruses that may fall into more than one of these categories.
- **Cavity (Space filler) Viruses –**
Cavity (Space filler) viruses attempt to maintain a constant file size when infecting a computer in order to try to avoid detection.
- **Tunneling Viruses –**
Tunneling viruses try to "tunnel" under anti-virus software while infecting.
- **Camouflage Viruses –**
Camouflage viruses attempt to appear as a benign program anti-virus scanners.
- **Virus Droppers –**
Virus Droppers are a special category of programs that place viruses on computers but are not by themselves an actual virus.

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Q. No. 5. How can protect and prevent computer from viruses?

Protecting your computer from viruses and other threats isn't difficult, but you have to be diligent. Here are some actions you can take:

1. Install an antivirus program.
Installing an antivirus program and keeping it up to date can help defend your computer against viruses. Antivirus programs scan for viruses trying to get into your email, operating system, or files. New viruses appear daily, so set your antivirus software to install updates automatically.
2. Don't open email attachments unless you're expecting them.
Many viruses are attached to email messages and will spread as soon as you open the email attachment. It's best not to open any attachment unless it's something expecting.

3. Keep your computer updated.
Virus updates that can help protect computer. Make sure that operating system receives these updates by turning automatic updating.
4. Use a firewall.
Firewall can help alert you to suspicious activity if a virus or worm attempts to connect to your computer. It can also block viruses, worms, and hackers from attempting to download potentially harmful programs to your computer.
5. Use your browser's privacy settings.
Being aware of how websites might use your private information is important to help prevent fraud and identity theft.
6. Turn on User Account Control (UAC).
When changes are going to be made to your computer that require administrator-level permission, UAC notifies you and gives you the opportunity to approve the change. UAC can help keep viruses from making unwanted changes.

Q. No. Define the Cyber crime Law in Pakistan?

Government of Pakistan proposed And Modified Cybercrime Bill 2015
(1) This Act may be called the Prevention of Electronic Crimes Act, 2015.
(2) It extends to the whole of Pakistan.
(3) It shall also apply to every citizen wherever he may be, and to every other person for the time being in Pakistan.

The cyber crime's bill has been passed by the National Assembly Standing Committee on IT and seems to be moving swiftly along.

There are 5 main points from summary of this new bill:

- It will be a crime to send text messages or photos to anyone's email address or phone without the recipient's consent.
- The police or FIA or any other agency won't need a warrant to search, seize or make arrests.
  Under sections 17 and 18 of the new bill, the political criticism and political expression in the form of analysis, commentary, blogs, cartoons, caricatures and memes has been criminalized. Authorities will decide what is moral and what is immoral.
- Under section 31, government can block or remove access to any website or online source if it deems it inappropriate.
- Under section 26, the ISPs, restaurants, malls, hotels, offices, airports bus stations and anywhere with Internet facility will be required to hold data record for 3 months.
- IT industry and the online social media users are terming the bill as senseless, draconian, punitive, arbitrary, and utterly far from reality.

Q. No. What is the Software Piracy and Law in Pakistan?

Syed Werasat Hussain Hashmi

Copyright law (Amendment) act of Pakistan 1992 Force law of 1962. Executing for piracy and also apply on computer piracy.

According to law the penalty up to 2,00,000 will apply the Violation of the law.



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